Finding the highest R^2 value in machine learning.

1.Multiple Linear Regression: R^2

Result : 0.7894790349867009

2.SVM Regression:R^2

kernel	**C**	**gamma**	**result**
Linear	1	----	-0.11166128719608448
Linear	1000	-------	0.634036931263208
Linear	1000	scale	0.7649311738597411
Sigmoid	1000	Scale	0.2874706948697686
Poly	1000	Scale	0.8566487675946576
Poly	1000	-----	0.8566487675946576
Sigmoid	1000	-----	0.2874706948697686

3.Decision Tree Regression:R^2

Criterion	**Splitter**	**Result**
Poisson	----	0.738082807313595
Poisson	Best	0.7311910176307761
Poisson	Random	0.7218112738774256
squared_error	Random	0.6912279756976367
absolute_error	Random	0.7442812244609339
absolute_error	best	0.6588379716627895
squared_error	best	0.6941323498581156
squared_error	------	0.6891305386521278
absolute_error	-----	0.6646951684456772

4.Random Forest Regression:R^2

n_estimators	random_state	Criterion	Result
100	0	Poisson	0.8535470331924936
100	0	squared_error	0.855098462902106
100	0	absolute_error	0.851521369792227
50	0	absolute_error	0.8503517423918491
50	0	squared_error	0.8510518652296519
50	0	Poisson	0.849996230439807
50	42	Poisson	0.8593601155052777
100	42	Poisson	0.856838131126881
100	42	squared_error	0.8550467930521874
100	42	absolute_error	0.848772742322136
100	100	absolute_error	0.8549840530550348
100	100	squared_error	0.8551183844308657
100	100	Poisson	0.8579794060835944

The final machine learning value in **R2score** = 0.8593601155052777

Winner = Random Forest Regression